DGMS 2019 • SECTION D • QUESTION 26

Mantle Reservoirs & Isotopic Geochemistry: The Bulk Silicate Earth (BSE) and Chondritic Uniform Reservoir (CHUR)

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Standard Petrology & Geochemistry Reference Guide

Question Statement

Q.26. Which one of the following geochemical reservoirs is defined by the Nd-isotopic composition of chondritic meteorites?

a (a) DM (Depleted Mantle)

b **(b) BSE (Bulk Silicate Earth)**

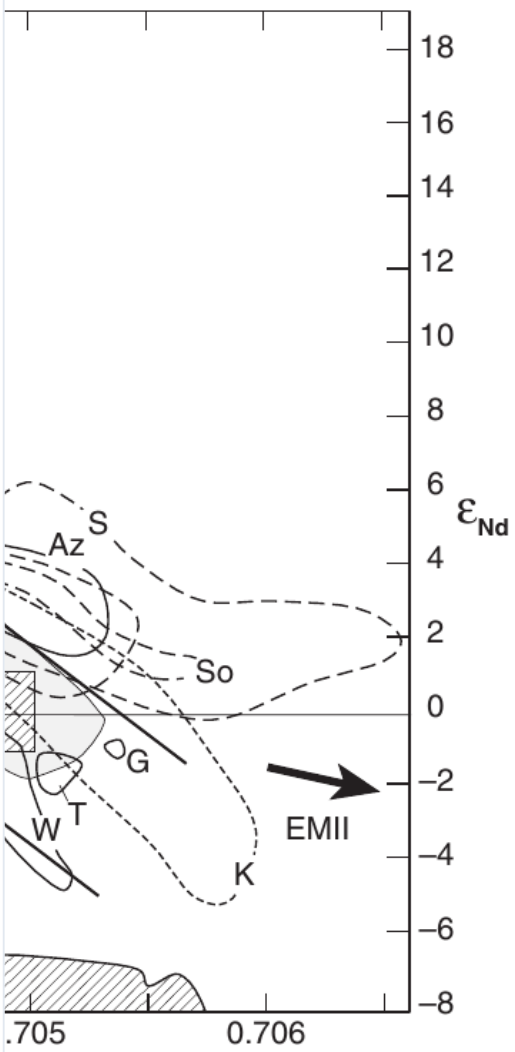
c (c) PREMA (Prevalent Mantle)

d (d) HIMU (High-μ Mantle)

Correct Answer: (b) BSE (Bulk Silicate Earth)

Official Textbook Figures & Mantle Reservoir Diagrams

has no distinctive end-member signature.



Key

As	Ascension
Az	Azores
B	Bouvet
C	Canary Is.
CI	Central Indian*
E	Easter Is.
EPR	East Pacific Rise
G	Gough
Gl	Galapagos
Gu	Guadaloupe
H	Hawaii
Ic	Iceland
IOR	Indian Ocean Ridge
K	Kerguelen
KH	Koolau, Hawaii
M	Marqueseas
MAR	Mid-Atlantic Ridge
S	Samoa
SH	St. Helena
So	Society Is.
T	Tristan da Cunha
Tu	Tubaii
W	Walvis

* includes: Amsterdam, Crozet, Marion-Prince Edward, Mauritius, Reunion, and Rodriguez.

BSE	Bulk Silicate Earth
EMI	Enriched Mantle I
EMII	Enriched Mantle II
DM	Depleted Mantle
HIMU	High μ Mantle
PREMA	Prevalent Mantle

:luding MORB and OIB, after Zindler and Hart (1986), Staudigel
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Figure 1: Winter (2010), Chapter 14, Figure 14-8. $^{143}Nd/^{144}Nd$ versus $^{87}Sr/^{86}Sr$ systematics for ocean basalts (MORB and OIB), showing the mantle reservoirs proposed by Zindler and Hart (1986). Notice the exact central placement of **BSE (Bulk Silicate Earth / bulk Earth)** at $\varepsilon_{Nd} = 0$ ($^{143}Nd/^{144}Nd = 0.512638$) and $^{87}Sr/^{86}Sr \approx 0.7045–0.7050$.

TABLE 5 Approximate Isotopic Ratios of Various Mantle Reservoirs

Reservoir	$^{87}Sr/^{86}Sr$	$^{144}Nd/^{143}Nd$	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{204}Pb$	$^{3}He/^{4}He$	$^{187}Os/^{186}Os$	$^{18}O/^{16}O$
End-Member Mantle							
DM	0.7015–0.7025	0.5133–0.5136	15.5–17.7	<15.45	7–9 R_A	0.123–0.126	low
HIMU	0.7025–0.7035	0.511–0.5121	21.2–21.7	15.8–15.9	2–6 R_A?	0.15	low
EMI	c 0.705	<0.5112	17.6–17.7	15.46–15.49	2–6 R_A?	0.152	low
EMII	>0.722	0.511–0.512	16.3–17.3	15.4–15.5	2–6 R_A?	0.156?	high
FOZO*	0.7030–0.7040	0.51280–0.51300	18.50–19.50	15.55–15.65	8–32 R_A	low	low
Other Mantle							
BSE	0.7052	0.51264	18.4	15.58	40–80 R_A?	0.129	low
PREMA	0.7033	<0.5128	18.2–18.5	15.4–15.5			
PHEM	0.7042–0.7052	0.51265–0.51280	18.4–19.0	15.5–15.6	>35 R_A	low	low
C	0.703–0.704	0.5128–0.5129	19.2–19.8	15.55–15.65	20–25+?	low	low
Continental Crust**	0.72–0.74	0.507–0.513	up to 28	up to 20	~0.1 R_A	high	high

*FOZO or some low-^{4}He reservoir, such as PHEM or C.

**Continental crust may be recycled back into the mantle via subduction.

DM, HIMU, EMI, EMII, and BSE from Rollinson (1993), pp. 233–236; PHEM from Farley et al. (1992); FOZO from Hauri et al. (1994); C from Hanan and

Figure 2: Winter (2010), Chapter 14, Table 14-5. Approximate Isotopic Ratios of Various Mantle Reservoirs. Note the entry for **BSE**: $^{143}Nd/^{144}Nd$ = **0.51264** (the exact chondritic meteorite value), $^{87}Sr/^{86}Sr$ = 0.7052, $^{206}Pb/^{204}Pb$ = 18.4.

The Earth as a whole has a $^{147}Sm/^{144}Nd$ ratio identical to that of chondrites (0.1967), and the initial $^{143}Nd/^{144}Nd$ of the Earth is also the same as that of chondrites. Thus the bulk Earth evolves along the same line as chondrites. Because Nd is more incompatible than Sm and will therefore be more enriched in a partial melt than Sm, the crust, formed by partial melting of the mantle, has a low Sm/Nd ratio and evolves along a line of lower slope than the bulk Earth. Partial melting depletes the mantle more in Nd than in Sm, so the mantle is left with a high Sm/Nd ratio and evolves along a steeper slope.

By replacing $^{143}Nd/^{144}Nd$ with ε_{Nd} we do not change the fundamental features of the isotope evolution diagram, but the slopes change (Figure 8.9b). Because at any time, ε_{Nd} is the deviation from chondritic $^{143}Nd/^{144}Nd$, the bulk Earth, as well as chondritic meteorites always have $\varepsilon_{Nd} = 0$. Thus they evolve along a line with slope 0. Crustal rocks, with low Sm/Nd, evolve along negative slopes and the mantle, with high Sm/Nd, evolves along a positive slope.

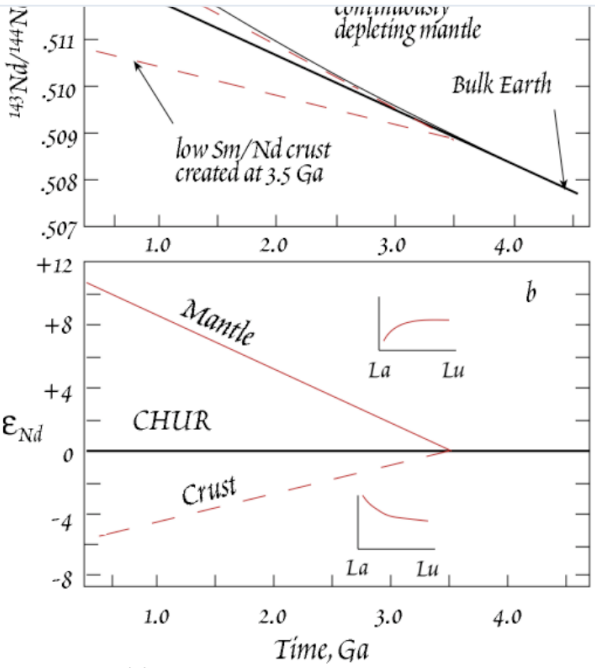


Figure 8.9. (a) Nd isotope evolution in mantle and crust. Bold line shows the evolution of the bulk earth or CHUR (chondritic uniform reservoir); also shown is the evolution of crust created at 3.5 Ga, the corresponding residual mantle, and the evolution of a continuously depleted mantle. (b) Evolution of bulk earth, crust, and mantle when $^{143}Nd/^{144}Nd$ is transformed to ε_{Nd}.

Figure 3: William M. White (2005), Chapter 8, Figure 8.9. (a) Nd isotope evolution in mantle and crust over 4.56 Ga. The bold horizontal reference line demonstrates the evolution of the **bulk earth or CHUR (chondritic uniform reservoir)**. (b) Evolution of bulk earth, crust, and mantle in ε_{Nd} notation, where Bulk Earth/CHUR is locked permanently at $\varepsilon_{Nd} \equiv 0$.

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os tend to be anti-correlated in the mantle and in man- sometimes called the "mantle array". The inverse cor- e parent-daughter pairs. Rb is less compatible than Sr, gmatic processes affect the Rb/Sr and Sm/Nd ratios in rse relationship between $^{87}Sr/^{86}Sr$ and ε_{Nd}. The anti- ı the mantle reflects the dominance of magmatic proc- Most of the mantle has ε_{Nd} higher and $^{87}Sr/^{86}Sr$ lower ı/Nd ratios have been high and Rb/Sr ratios have been)RB) tend to have lowest $^{87}Sr/^{86}Sr$ and highest ε_{Nd}. This RB is derived must be highly depleted in incompatible partial melting.

$^{87}Sr/^{86}Sr$ and lower ε_{Nd} than the bulk Earth. The anti- eaker in crustal rocks. In part, this reflects the impor- d in evolution of the crust. It also reflects a weaker in- ous igneous rocks. The lower continental crust appears upper continental crust, but it should be emphasized osition between the two, and that lower crustal rocks ontinental basalts range to much higher $^{87}Sr/^{86}Sr$ and

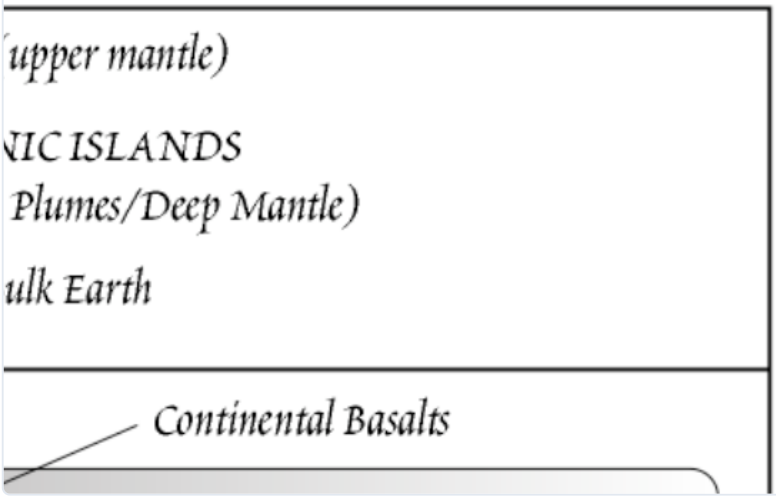


Figure 4: William M. White (2005), Chapter 8, Figure 8.11. Sr and Nd isotope ratios in major geochemical reservoirs. The composition of the **bulk Earth (BSE)** lies exactly at the intersection of the horizontal datum line at $\varepsilon_{Nd} = 0$ and $^{87}Sr/^{86}Sr \approx 0.705$.

In-Depth Option Evaluation & Geochemical Breakdown

(b) BSE (Bulk Silicate Earth) — [CORRECT]

Definition: The Bulk Silicate Earth (BSE), also termed the *Primitive Mantle (PUM)*, represents the bulk composition of the Earth's silicate portion (crust + mantle) prior to any differentiation into continental crust and residual depleted mantle, but *after* core formation.

Why Chondritic Nd Isotopes Define BSE: Both Samarium (Sm) and Neodymium (Nd) are **Refractory Lithophile Elements (RLE)**.

- *Refractory:* They possess extremely high condensation temperatures in the solar nebula (> 1400 K), meaning they did not volatilize during planetary accretion.
- *Lithophile:* They do not dissolve in metallic iron and thus did not enter the Earth's core during planetary core segregation.

Because neither element was fractionated by volatility nor core separation, the whole silicate Earth **must have the exact same Sm/Nd ratio as chondritic meteorites**. Consequently, the Nd-isotopic evolution of the Bulk Silicate Earth is modeled by the **Chondritic Uniform Reservoir (CHUR)** (DePaolo & Wasserburg, 1976). Today, CHUR and BSE have an identical $^{143}Nd/^{144}Nd$ value of **0.512638**, defining $\boldsymbol{\varepsilon_{Nd} \equiv 0}$!

(a) DM (Depleted Mantle / DMM) — [INCORRECT]

Characteristics: Depleted Mantle (DM) is the residue left behind in the upper mantle after the extraction of continental crust over geological time. It is the source reservoir for normal mid-ocean ridge basalts (N-MORB).

Isotopic Signature: During partial melting of the primitive mantle, Nd is more incompatible than Sm ($D_{Nd} < D_{Sm}$), so Nd partitioned preferentially into the crustal melt. This left the residual mantle (DM) with a **higher Sm/Nd ratio** than chondrites. Over 2–3 billion years of ^{147}Sm decay, DM evolved to **higher $^{143}Nd/^{144}Nd$ ratios (0.5131–0.5136)**, corresponding to strongly positive ε_{Nd} values (+10 to +12). Hence, DM is distinctly *non-chondritic*!

(c) PREMA (Prevalent Mantle) — [INCORRECT]

Characteristics: Proposed by Zindler and Hart (1986), PREMA represents a common focal component toward which many oceanic island basalt (OIB) arrays converge from different geographic sectors.

Isotopic Signature: PREMA has $^{143}Nd/^{144}Nd \approx 0.5128–0.5130$ ($\varepsilon_{Nd} \approx +4$ to $+6$), intermediate between DM and BSE, and $^{87}Sr/^{86}Sr \approx 0.7033$. It represents a mildly depleted, well-mixed mantle component (possibly lower mantle or entrained plume material), not the chondritic reservoir.

(d) HIMU (High-μ Mantle) — [INCORRECT]

Characteristics: HIMU derives its name from "high μ" (where $\mu = {}^{238}U/{}^{204}Pb$). It was created by ancient subduction and mantle storage (~1.5–2.0 Ga) of recycled altered oceanic crust (basaltic slabs) that lost Pb during subduction dehydration, resulting in an exceptionally elevated U/Pb ratio.

Isotopic Signature: Characterized by extremely radiogenic lead isotopes ($^{206}Pb/^{204}Pb > 20.5–21.5$), very low $^{87}Sr/^{86}Sr$ (~0.7028), and intermediate $^{143}Nd/^{144}Nd$ (0.5128–0.5129, $\varepsilon_{Nd} \approx +4$). It has no genetic or isotopic equivalence to chondritic meteorites.

Theoretical Foundations: The Sm-Nd Isotopic System & CHUR

1. Radioactive Decay Scheme

The Sm-Nd isotopic system is based on the alpha decay of ^{147}Sm to ^{143}Nd:

Sm-Nd Alpha Decay Law:

$^{147}Sm \rightarrow {}^{143}Nd + \alpha$ (decay constant $\lambda = 6.54 \times 10^{-12}$ yr^{-1}, half-life $t_{1/2} = 106$ Ga)

$$\left(\frac{^{143}Nd}{^{144}Nd}\right)_{present} = \left(\frac{^{143}Nd}{^{144}Nd}\right)_{initial} + \left(\frac{^{147}Sm}{^{144}Nd}\right) \times (e^{\lambda t} - 1)$$

2. The CHUR (Chondritic Uniform Reservoir) Concept

In 1976, Donald DePaolo and Gerald Wasserburg introduced the **Chondritic Uniform Reservoir (CHUR)** model. They analyzed carbonaceous and ordinary chondritic meteorites and demonstrated that their Sm/Nd ratios are remarkably uniform:

Present-Day CHUR Standard Parameters:

$(^{147}Sm / ^{144}Nd)_{CHUR}^{0} = 0.1967$

$(^{143}Nd / ^{144}Nd)_{CHUR}^{0} = 0.512638$ (normalized to $^{146}Nd/^{144}Nd = 0.7219$)

3. The Epsilon Neodymium (ε_{Nd}) Notation

Because differences in $^{143}Nd/^{144}Nd$ occur only in the 4th and 5th decimal places, DePaolo and Wasserburg defined parts per 10,000 deviations from CHUR/BSE:

Epsilon Nd Equation:

$$\varepsilon_{Nd} = \left[\left((^{143}Nd/^{144}Nd)_{sample} / (^{143}Nd/^{144}Nd)_{CHUR} \right) - 1 \right] \times 10{,}000$$

- If $\varepsilon_{Nd} = 0$ → Sample is identical to Bulk Silicate Earth (BSE / CHUR).
- If $\varepsilon_{Nd} > 0$ → Sourced from Depleted Mantle (DM), higher Sm/Nd ratio (LREE-depleted).
- If $\varepsilon_{Nd} < 0$ → Sourced from Enriched Reservoir / Continental Crust, lower Sm/Nd ratio (LREE-enriched).

Comprehensive Comparison of Zindler & Hart (1986) Mantle Reservoirs

Reservoir	Full Name & Origin	$^{143}Nd/^{144}Nd$	ε_{Nd}	$^{87}Sr/^{86}Sr$	$^{206}Pb/^{204}Pb$	Chondritic Relationship
BSE	**Bulk Silicate Earth (Primitive Mantle / PUM)**	**0.512638**	**0**	**0.7045–0.7052**	**18.40**	**Identical to Chondritic Meteorites (CHUR)**
DM (DMM)	**Depleted Mantle** (MORB source, crust residue)	0.5131–0.5136	+10 to +12	0.7015–0.7025	15.5–17.7	Non-chondritic (High Sm/Nd, depleted in Nd)
EM-I	**Enriched Mantle I** (Subducted lower crust / pelagic sed)	< 0.5112–0.5124	-12 to 0	0.7045–0.7055	17.6–17.8	Non-chondritic (Low Sm/Nd, enriched in Nd)
EM-II	**Enriched Mantle II** (Subducted upper crust / terrigenous sed)	0.5110–0.5122	-15 to -2	> 0.707–0.722	16.3–17.3	Non-chondritic (Very high Rb/Sr, crustal)
HIMU	**High-μ Mantle** (Recycled dehydrated oceanic crust)	0.5128–0.5129	+4 to +5	0.7028–0.7030	> 20.5–21.8	Non-chondritic (Extreme U/Pb fractionation)
PREMA	**Prevalent Mantle** (Common focal mixing component)	0.5128–0.5130	+4 to +6	0.7033–0.7035	18.2–18.5	Non-chondritic (Intermediate depleted)
FOZO	**Focus Zone** (Deep primitive/ plume mantle, high $^{3}He/^{4}He$)	0.5128–0.5130	+4 to +6	0.7030–0.7040	18.5–19.5	Non-chondritic (Deep lower mantle component)

Direct Textbook Quotations & Citations

John D. Winter (2010), *Principles of Igneous and Metamorphic Petrology, Chapter 14 (Oceanic Intraplate Volcanism), p. 275:*

"A second reservoir may be a non-depleted mantle, called **BSE (bulk silicate Earth)** *by Zindler and Hart (1986) and primitive upper mantle (PUM) or primary uniform reservoir by others.* **BSE reflects the isotopic signature of the primitive mantle as it would evolve to the present without any subsequent fractionation of radioactive elements** *(which presumably were extracted to form continental and oceanic crust)... In Table 14-5, BSE has* $^{143}Nd/^{144}Nd$ = **0.51264** *and* $^{87}Sr/^{86}Sr$ = *0.7052."*

— **Winter (2010), p. 275 & Table 14-5**

William M. White (2005), *Geochemistry, Chapter 8 (Radiogenic Isotope Geochemistry), p. 332:*

"The Earth as a whole has a $^{147}Sm/^{144}Nd$ *ratio identical to that of chondrites (0.1967), and the initial* $^{143}Nd/^{144}Nd$ *of the Earth is also the same as that of chondrites.* **Thus the bulk Earth evolves along the same line as chondrites.** *Because at any time,* ε_{Nd} *is the deviation from chondritic* $^{143}Nd/^{144}Nd$, **the bulk Earth, as well as chondritic meteorites always have** $\varepsilon_{Nd} = 0$*."*

— **White (2005), p. 332**

Kula C. Misra (2012), *Introduction to Geochemistry: Principles and Applications, Chapter 12 (Radiogenic Isotopes), p. 312:*

"The concept of CHUR (Chondritic Uniform Reservoir) is based on the assumption that the Earth's mantle has evolved with a Sm/Nd ratio equal to that of chondritic meteorites... **The present-day** $^{143}Nd/^{144}Nd$ **ratio of CHUR (0.512638) defines the Bulk Silicate Earth (BSE)**."

— **Misra (2012), p. 312**

Exam Traps & Rapid Elimination Strategy

Rule of Thumb for UPSC Geo-Scientist & DGMS:

1. Whenever a question connects **chondritic meteorites** / **CHUR** to a mantle reservoir, immediately look for **BSE (Bulk Silicate Earth)** or **PUM (Primitive Upper Mantle)**. They are synonymous in isotope geochemistry.

2. **DM (Depleted Mantle)** is always associated with **N-MORB**, high positive ε_{Nd} (+10 to +12), and low $^{87}Sr/^{86}Sr$.
3. **HIMU** is always associated with **high $^{206}Pb/^{204}Pb$ (> 20.5)** and recycled oceanic crust.
4. **EM-I** / **EM-II** are always associated with **negative ε_{Nd}**, high $^{87}Sr/^{86}Sr$, and recycled continental crust / sediments.
5. **PREMA** is the common convergence point of OIB mixing arrays ($\varepsilon_{Nd} \approx +4$ to +6).